

RS2000 hardware user guide

Below is a practical **RS2000 hardware user guide** for the **Impinj Indy RS2000 UHF RFID reader module**. The exact pinout and PCB footprint must still be checked against your exact module/datasheet, because “RS2000/R2000 module” is also used by some Chinese module vendors.

1. What RS2000 is

The **Indy RS2000** is an embedded **RAIN RFID / UHF RFID reader module** based on the Impinj R2000 reader chip. It supports **EPCglobal UHF Class 1 Gen2 / ISO 18000-63**, formerly called ISO 18000-6C. The official datasheet describes it as a turnkey embedded reader module requiring power, host communication, and antenna connections.

Official PDF sources:

Document	Link
Indy RS2000 Datasheet	
Indy RS2000 Development Kit Quick Start Guide	

The public datasheet references an **RS2000 Hardware User's Guide**, but I found the datasheet and development-kit quick-start PDF publicly; the full hardware user guide itself may require Impinj support/customer access.

RS2000 Hardware User Guide

2. Main hardware connections

A basic RS2000 embedded system needs:

Connection	Purpose
VDC_IN	Main DC power input
GND	Ground
ANT1 / ANT2 / ANT3 / ANT4	UHF RFID antenna ports

UART1 TX/RX	Main host communication
ENABLE	Enables/disables the module
WKUP	Wake-up control, optional/recommended depending design
GPIOs	General-purpose input/output
UART2 TX/RX	Debug interface
STATUS / HEALTH	Test/status signals

The datasheet block diagram shows **VDC_IN + GND**, at least one antenna port, **UART1 TX/RX**, and **ENABLE** as required connections. UART2 debug, status/health, wake-up, and GPIOs are recommended or optional depending on the design.

3. Power supply

The official RS2000 datasheet lists the DC power supply range as:

Parameter	Value
Supply voltage	3.2 V to 5.25 V DC

So your host board should provide a clean DC supply in that range.

Recommended design notes:

Item	Recommendation
Power current capacity	Use a regulator with enough margin for RF transmit peaks
Ground	Use a solid ground plane
Decoupling	Place capacitors close to the module power pins
Noise	Keep switching regulator noise away from RF path

Enable pin	Drive high to enable the module; drive/toggle low for disable/reset behavior
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Important: RFID modules can draw much more current during RF transmission than when idle. Do not test it from a weak USB-to-serial adapter power pin.

4. Host interface

The normal control interface is **UART1 TX/RX**.

Typical connection:

RS2000 side	Host MCU/CPU side
RS2000 TX	Host RX
RS2000 RX	Host TX
GND	Host GND

Do not connect TX-to-TX or RX-to-RX. Cross them.

The datasheet notes that host communication speed can limit tag read rate, and says the default baud rate of **115,200 baud** can limit tag read rate to about **300 tags/second**.

For MCU use, check the logic level. Many RFID modules use **3.3 V UART logic**, not 5 V TTL. If your MCU is 5 V, use level shifting.

5. Antenna connection

The RS2000 supports up to **4 antenna ports total**, with some connectorized and some surface-mount ports depending on the module/SKU. The datasheet states that only one antenna is required for RFID operation, but up to four may be used.

Antenna design checklist:

Item	Recommendation
Frequency	Match region: 865–868 MHz EU or 902–928 MHz US
Impedance	Usually 50 Ω antenna system

Cable	Use low-loss coax
Connector	Match the module connector type
Antenna gain	Must comply with local RF regulations
Placement	Keep antenna away from metal unless using a metal-mount design

For testing, start with one known-good UHF RFID antenna on ANT1.

6. Enable and reset behavior

The datasheet block diagram says **ENABLE** is used to enable or disable the part. If driven low, the internal power supplies are disabled. The module can also be reset by toggling ENABLE; the pin must be driven high to enable the part.

Basic behavior:

ENABLE state	Result
High	Module enabled
Low	Module disabled / powered down internally
Toggle low then high	Reset-like behavior

For your board, do not leave ENABLE floating. Pull it to a known state.

7. GPIO, status, health, wake-up

The RS2000 provides additional control/status pins, but these are not all mandatory for a simple reader.

Signal	Use
GPIO	Application-specific input/output
STATUS / HEALTH	Debug or module health monitoring
WKUP	Wake-up / low-power control
UART2	Debug output / troubleshooting

For a first prototype, you can usually start with:

1	Power
2	Ground
3	UART1 TX/RX
4	ENABLE
5	One antenna

Then add GPIO/status/wake-up later.

8. Mechanical mounting

The official datasheet lists the module dimensions as:

Parameter	Value
Dimensions	38 mm × 52 mm × 4 mm
Package type	35-pin surface-mount package
Mass	About 12 g

The module is intended to be attached to a PCB using SMT. The datasheet also mentions a screw hole that can help thermal conduction to a chassis or heatsink.

Hardware layout advice:

Area	Recommendation
Ground	Large ground plane under/near module
RF traces	50 Ω controlled impedance
Thermal	Provide copper area or chassis contact
Keepout	Keep noisy digital/switching circuits away from RF area
Mounting	Follow official land pattern exactly

9. RF output power

The datasheet gives RS2000 Tx output power as:

Parameter	Value
Tx output power range	+10 to +31.5 dBm
Default	+30 dBm
Japan max noted	+30 dBm

Practical meaning:

Power	Meaning
10 dBm	Low power, short read range
20 dBm	Medium test power
30 dBm	High power, long range
31.5 dBm	Maximum module output, where allowed

Do not simply set maximum power everywhere. Use the lowest power that gives stable reads, especially near many tags.

10. Supported RFID protocol

The RS2000 supports:

Item	Support
Air interface	RAIN RFID / UHF RFID
Protocol	EPC Gen2 / ISO 18000-63
Older name	ISO 18000-6C
Features	Inventory, read, write, lock, kill, block permalock, QT

The datasheet lists Gen2 functionality including inventory features such as FastID, TagFocus, select, session, and target.

11. Basic bring-up procedure

Use this order when testing the hardware.

Step 1 — Visual inspection

Check:

- 1 Power polarity
- 2 No solder bridges
- 3 Antenna connected
- 4 UART TX/RX crossed correctly
- 5 ENABLE driven high
- 6 Ground shared between host and module

Step 2 — Power on

Measure:

- 1 VDC_IN = within 3.2-5.25 V
- 2 GND stable
- 3 No excessive current draw
- 4 Module does not overheat

Step 3 — UART communication

Connect host UART and open serial communication. Common starting value:

- 1 Baud rate: 115200
- 2 Data bits: 8
- 3 Parity: none
- 4 Stop bits: 1
- 5 Flow control: none

The exact protocol depends on the firmware/toolkit used by your module.

Step 4 — Set region

Set the regulatory region before transmitting:

- 1 US: 902-928 MHz
- 2 EU: 865-868 MHz
- 3 China/Korea/Japan: depends on local band

Do not transmit on the wrong frequency band.

Step 5 — Set RF power

Start low:

- 1 Power: 10-20 dBm for bench test
- 2 Then increase gradually

Step 6 — Inventory test

Place one known EPC Gen2 tag near the antenna and run inventory. After one tag works, test multiple tags.

12. Common problems and fixes

Problem	Likely cause	Fix
No UART response	TX/RX not crossed, wrong baud, ENABLE low	Check wiring, baud, enable
Module resets	Weak power supply	Use stronger regulator, add decoupling
No tag reads	Wrong region, no antenna, bad antenna cable, low power	Check RF settings and antenna
Very short read range	Low power, poor antenna, metal nearby	Increase power, reposition antenna
Reads unstable	RF noise, bad ground, reflections	Improve grounding and antenna placement
Module overheats	High duty cycle, poor thermal path	Add copper/heatsink, reduce duty cycle
Works in demo board but not custom PCB	Bad footprint/RF layout/power	Compare against reference design

13. Minimum prototype wiring

For a simple MCU/PC-controlled prototype:

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1 RS2000 VDC_IN  → 3.3 V or 5 V regulated supply within allowed range
2 RS2000 GND     → Host GND
3 RS2000 UART1_TX → Host UART_RX
4 RS2000 UART1_RX → Host UART_TX
5 RS2000 ENABLE  → Host GPIO or pull-up to enable
6 RS2000 ANT1    → UHF RFID antenna
```

That is the minimum hardware set to start reading tags.

14. Important warning

For a real product, you must check:

- 1 Regional RF certification
- 2 Antenna gain limit
- 3 Output power limit
- 4 FCC / CE / ETSI / local radio rules
- 5 Labeling requirements
- 6 Thermal performance
- 7 ESD protection

The RS2000 datasheet mentions FCC and Canada modular certification for specific configurations, and ETSI behavior separately. That means your final product design still needs regulatory review.

could you give me the download link for the user guide pdf?

I'll look specifically for the RS2000 **Hardware User Guide PDF**, not just the datasheet/quick-start, because vendors often list those separately and hide the real guide behind support portals.

Thought for 4s

I found **public PDFs for the datasheet and dev-kit quick start**, but I did **not find a public direct PDF download** for the official **Impinj Indy RS2000 Hardware User's Guide**.

The RS2000 dev-kit quick-start says the **Datasheet, RS2000 Hardware User Guide, and RS2000 IRI Toolkit User Guide** are available at **support.impinj.com**, which likely requires an Impinj support account/login.

Public download links I found

Document	PDF download link
Indy RS2000 Datasheet	https://rfid.atlasrfidstore.com/hubfs/1_Tech_Spec_Sheets/Impinj/ATLAS%20Impinj%20Indy%20RS2000%20RFID%20Reader%20Module%20Data%20Sheet.pdf
Indy RS2000 Development Kit Quick Start Guide	https://rfid.atlasrfidstore.com/hubfs/1_Tech_Spec_Sheets/Impinj/ATLAS%20Impinj%20Indy%20RS2000%20RFID%20Reader%20Module%20Development%20Kit%20Quick%20Start%20Guide.pdf

Closest public hardware guide

I also found a public **MTI M200-RS Hardware User's Guide**, which says the M200-RS is **pin-to-pin, 100% compatible with Impinj Indy RS2000**. This is not the official RS2000 guide, but it may be very useful for hardware pins, UART, power, antenna, and dev-board details.

Document	Link
MTI M200-RS Hardware User's Guide — RS2000 compatible	https://www.manualslib.com/manual/1978132/Mti-M200-Rs.html

For the **official RS2000 Hardware User Guide PDF**, the best source appears to be Impinj Support: **support.impinj.com**.